

A2 Warm Up 7-4 Properties of Logs

Before you begin, please write your 3 rules that work backwards and forwards (see your notes)

1. Condense the logarithm:

$$3 \log y + b \log c$$

$$\log y^3 + \log c^b$$

$$\boxed{\log(y^3 \cdot c^b)}$$

2. Condense the logarithm:

$$2 \log g - d \log q$$

$$\log g^2 - \log q^d$$

$$\boxed{\log\left(\frac{g^2}{q^d}\right)}$$

NOT $\frac{\log g^2}{\log q^d}$

3. Expand the logarithm fully using the properties of logs.

Express the final answer in terms of $\log x$.

$$\log 5x^3$$

multiply $\Rightarrow$ product

$$\log 5 + \log x^3$$

Remember: expand

$$\boxed{\log 5 + 3 \log x}$$

4. Expand the logarithm fully using the properties of logs.

Express the final answer in terms of $\log x$, and $\log y$.

$$\log \frac{x^3}{y^2}$$

$$\log x^3 - \log y^2$$

$$\boxed{3 \log x - 2 \log y}$$

5. Expand the logarithm fully using the properties of logs.

Express the final answer in terms of $\log x$, $\log y$, and $\log z$.

$$\log \frac{\sqrt[3]{y^5}}{z^2 x^5}$$

Rewrite

$$= \frac{\log y^{\frac{5}{3}}}{\log z^2 \cdot x^5}$$

power
root

$$= \log y^{\frac{5}{3}} - (\log z^2 + \log x^5)$$

$$= \log y^{\frac{5}{3}} - \log z^2 - \log x^5$$

$$\boxed{\frac{5}{3} \log y - 2 \log z - 5 \log x}$$